Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *MILU* | ***Context:*** *Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output form mismatch is fundamental and acknowledged by the paper itself. MILU evaluates via log-likelihood option selection [Q52, Q53, Q54, Q55] or zero-shot JSON-formatted option choice for API models [Q56, Q57]. The authors note explicitly that this 'may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting' [Q85]. The deployment requires open-ended Hindi-language explanatory generation — a fundamentally different output form. Dataset analysis confirms the schema contains only a 'target' MCQ label field [DATASET-D1, D9] with no infrastructure for evaluating fluency, correctness, or pedagogical quality of free-text Hindi output. External evidence confirms no comparable Hindi-language NLG benchmark for exam rationale exists [WEB-13, WEB-14].

ID	Evidence
Q52	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q53	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q54	“"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., logP(a\” (p.5)
Q55	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q56	“The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities.” (p.5)
Q57	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? … 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest … what is the annual rate of interest? … 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition

Strengths

- MCQ accuracy can validate the verdict-correctness portion of the deployment's output.
- Per-language, per-subject accuracy reporting [Q59, Q67] supports diagnostic identification of weak areas.
- Multiple shot configurations (0/1/5-shot) [Q50] provide robustness checks on the verdict accuracy.

ID	Evidence
Q59	“We report 5-shot accuracies for all open models (except for LLAMA-3.1-70B-INSTRUCT & LLAMA-3.1-405B for which we report 0-shot accuracy) with the accuracy averaged across all the domains per language.” (p.6)
Q67	“Table 4: Evaluation results of all the language-specific fine-tuned models on MILU. We report domain level 5-shot accuracies for all the models on the language supported by the model. Higher values indicate better model performance for the given domain.” (p.7)

Q50	“All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups.” (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Major mismatch: MILU outputs MCQ option labels [Q55, Q57]; deployment requires open-ended Hindi rationale plus verdict [elicitation Q3]. The verdict portion is partially covered; the rationale portion is wholly uncovered.

OF-2: Not applicable — deployment is text-only, no TTS requirement.

OF-3: Target population is graduate-level (high literacy within cohort), so accessibility is not the bottleneck — the bottleneck is rationale quality, which MILU cannot evaluate.

OF-4: Form mismatches: (a) MCQ vs. open-ended generation; (b) log-likelihood scoring [Q53] vs. free-text Hindi generation quality; (c) no fluency/coherence/pedagogical metrics. The paper itself flags this limitation [Q85].

ID	Evidence
Q55	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q57	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q3	“We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%.” (p.1)
Q53	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)

Requires Expert Verification

- A Hindi-language NLG evaluation specialist should design a custom rationale-quality rubric (factual accuracy, fluency, pedagogical clarity, register) since no off-the-shelf Hindi resource exists [WEB-13, WEB-14].

Remediation

Gap 1: MILU evaluates only MCQ option selection; deployment requires open-ended Hindi rationale generation.

Recommendation: Design and operationalize a separate Hindi rationale-quality benchmark with (a) human-rated rubric (factual accuracy, fluency, pedagogical clarity, register), (b) ~500–1000 expert-authored gold rationales for a stratified sample of MILU Hindi items, (c) automated metrics (BLEURT/COMET-style for Hindi, plus LLM-as-judge calibrated against expert ratings).